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ORIGINAL PAPER

Arousal level and exemplar variability of emotional face and voice encoding influence expression-independent identity recognition

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Abstract

Emotional stimuli and events are better and more easily remembered than neutral ones. However, this advantage appears to come at a cost, namely a decreased accuracy for peripheral, emotion-irrelevant details. There is some evidence, particularly in the visual modality, that this trade-off also applies to emotional expressions, leading to a difficulty in identifying an unfamiliar individual's identity when presented with an expression different from the one encountered at encoding. On the other hand, past research also suggests that identity recognition memory benefits from exposure to different encoding exemplars, although whether this is also the case for emotional expressions, particularly voices, remains unknown. Here, we directly addressed these questions by conducting a series of voice and face identity memory online studies, using a within-subject old/new recognition test in separate unimodal modules. In the Main Study, half of the identities were encoded with four presentations of one single expression (angry, fearful, happy, or sad; *Uni* condition) and the other half with one presentation of each emotion (*Multi* condition); all identities, intermixed with an equal number of new ones, were presented with a neutral expression in a subsequent recognition test. Participants ($N = 547$, 481 female) were randomly assigned to one of four groups in which a different *Uni* single emotion was used. Results, using linear mixed models on response choice and drift-diffusion-model parameters, revealed that high-arousal expressions interfered with emotion-independent identity recognition accuracy, but that such deficit could be compensated by presenting the same individual with various expressions (i.e., high exemplar variability). These findings were confirmed by a significant correlation between memory performance and stimulus arousal, across modalities and emotions, and by two follow-up studies (Study 1: $N = 172$, 150 female; Study 2: $N = 174$, 154 female), which extended the original observations and ruled out some potential confounding effects. Taken together, the findings reported here expand and refine our current knowledge of the influence of emotion on memory, and highlight the importance of, and interaction between, exemplar variability and emotional arousal in identity recognition memory.

Keywords Emotional Expression · Arousal · Exemplar Variance · Face Recognition · Voice Recognition

Introduction

Most people will immediately recognize the woman (and her scream) in Hitchcock's *Psycho* shower murder scene, even those who didn't see the actual movie. Consistent with this anecdotal observation, a large body of research has showed a superior memory recognition accuracy for emotionally expressed faces (e.g., Sergerie et al., 2005; LaBar & Cabeza, 2006) and voices (e.g., Armony et al., 2007; Aubé et al., 2013; Pichora-Fuller et al., 2016), compared to neutral ones. This emotional memory advantage has also been reported for other emotional stimuli, such as objects, scenes, and events (e.g., Cahill et al., 1994; Kensinger, 2004; Kensinger & Schacter, 2005; Righi et al., 2012). Although the

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exact mechanisms of this emotional enhancement of episodic memory are still unclear, emotional arousal has been regarded as an important factor in influencing emotional memory (e.g., Mather & Sutherland, 2011; Ack Baraly et al., 2016; Cahill et al., 1994). When an experience triggers an arousal response, memory formation stages — including encoding, consolidation, and retrieval — can be affected, possibly by amygdala-mediated processes (McGaugh, 2004; Qasim et al., 2023), increasing the likelihood of the experience being remembered (for a review, see Kensinger, 2009).

However, this emotion-mediated memory enhancement may come at a cost, namely decreased memory accuracy for emotionally-irrelevant or "extrinsic" aspects of the stimulus (that is, spatially, temporally, or conceptually distinct from the emotion-specific information; Kensinger, 2009). This phenomenon, sometimes referred to as the central-peripheral trade-off (Kensinger, 2007; Mather & Sutherland, 2011), is particularly pronounced for high arousal events, possibly due to a narrowing of the attention focus (Kensinger, 2009). Indeed, stimuli signaling threat are processed in a more automatic fashion (Armony, Vuilleumier, Driver, & Dolan, 2001; Sander et al., 2005), likely capturing attention (Mogg & Bradley, 1999; Dolcos et al., 2020), and thus resulting in a more effective encoding process (Talmi et al., 2008). This may reflect the evolutionary advantage prioritizing survival over other goals and motivations (e.g., Unkelbach et al., 2020; Norris, 2021; Rozin & Royzman, 2001).

There is some evidence suggesting that the proposed trade-off also applies to emotional expressions; in that case, the memory enhancement for the previously encountered stimulus would be associated with a decrease in the recognition of the (expression-independent) individual's identity. Coming back to the *Psycho* example, how many of those who rapidly and without hesitation recognize the shower scene would be able to identify the actress (Janet Leigh) if they saw or heard her in another movie? Whereas this question has been less investigated, some studies have shown a decrease in identity recognition of individuals, especially for unfamiliar ones, when the emotional expression changes between encoding and retrieval (see Bruce, 1982, Liu et al., 2014, Nomi et al., 2013 for faces; Saslove & Yarmey, 1980, Stevenage & Neil, 2014, Xu & Armony, 2021 for voices).

On the other hand, a number of identity representation models, largely based on face perception — such as the averaging/prototype (e.g., Benson & Perrett, 1993; Burton et al., 2005) and pictorial coding models (e.g., Longmore et al., 2008) —, collectively propose that stimulus-independent person recognition may benefit from exposure to within-person exemplar variance (i.e., different instances of the same individual along a given feature dimension, such as view-point for faces or speaking style for voices), which allows for the formation of a more stable identity representation, or a greater likelihood of a match between a novel encounter and previously stored instances. Studies using uncontrolled within-person exemplar variance in learning of faces (ambient face images, e.g., Murphy et al., 2015; Ritchie & Burton, 2017; Matthews et al., 2018) and voices (Lavan et al., 2019a) provide support for this proposal.

Interestingly, there is some empirical data, though somewhat limited and not always consistent, suggesting a similar benefit after encoding exemplars in multiple emotional expressions of the same individual. Liu and colleagues (2015) examined this hypothesized advantage using static face images, by comparing face recognition in a novel emotional expression after they were encoded with a single exemplar (i.e., only one emotional expression) or with multiple exemplars (i.e., several different emotional expressions). Though the expression-independent face recognition was overall above-chance, the advantage of encoding multiple emotional expressions was only significant when compared to identities encoded with a single neutral expression. In their follow-up work using both static images and dynamic face videos (Liu et al., 2016), they found that exposure to single or multiple dynamic expressions led to a similar identity recognition presented with a new expression. In contrast, exposure to static faces resulted in a significantly weaker recognition of the new-expression identities. This is not surprising as research has shown that people are more efficient in learning and recognizing dynamic faces (e.g., O'Toole et al., 2002; Xiao et al., 2014). One noticeable feature in both studies is that the emotions in the exemplars encoded in both multiple-expression and single-expression conditions were randomly chosen from six basic emotions, rather than using fixed ones for each participant. Whereas this could provide more generalizable conclusions that are not emotion-specific, it may have also obscured possible emotion-dependent influences, which could have contributed to the small magnitude of the effects observed. Nonetheless, these results collectively point to an emotional expression advantage for face identity memory, albeit with a weaker magnitude compared to the effect obtained with ambient images.

The relation between emotional expressions and identity memory has been less studied for voices. Consistent with the central-peripheral trade-off hypothesis, Lavan and colleagues (2019b) found that learning high expressive unfamiliar voices interfered with identity categorization in a sorting task, compared to learning low expressive voices. Regarding exemplar variability, a recent study reported an advantage in recognition speed, though not in recognition accuracy, for speakers who were encoded with two (fearful

and happy), compared to one (fearful), prosodies (Xu & Armony, 2021).

In summary, emotional expressions, or at least some of them, seem to influence subsequent memory: single expressions lead to an enhancement of same-identity/same-emotion (i.e., same stimulus) recognition, but a reduction of same-identity/different-emotion accuracy. However, exposure to multiple emotional expressions may overcome the latter deficit, at least to some extent, by helping form a stable emotion-independent identity representation. Nonetheless, the putative mnemonic advantage of multiple emotional within-person exemplars in both face and voice has not been thoroughly tested. The limited number of studies, as mentioned above, did provide, mostly indirect, and sometimes conflicting, support for this multiple exemplar advantage. Moreover, it remains unclear whether such effects are specific to certain categories of emotion, such as high arousing ones (e.g., fear), or are generalizable to others. This is relevant as not every expression poses the same level of salience or recruits the same level of attentional resources (e.g., Lundqvist et al., 2015; Vuilleumier & Huang, 2009).

Here, we report three studies to directly test whether presentation of multiple emotions during encoding leads to a better identity recognition, compared to the repeated presentation of a single emotion and, if so, whether this effect depends on the specific emotions presented (see Table 1 for the expression combinations used in each study). We did so through a recognition memory task in which subjects encoded identities through speech clips or face images (in separate modules). Half of the identities were presented with four different emotions (*Multi* condition), while for the other half, a single emotion was presented four times (*Uni* condition). The specific emotion for the *Uni* condition was varied across subjects. Two additional studies were subsequently conducted to further clarify and help interpret the findings obtained in the main study. Choice (Older/Younger and New/Old during encoding and recognition, respectively; see Methods) and the corresponding response times (RTs), as well as response confidence during recognition, were collected. We analyzed recognition accuracy and an integrated accuracy-speed index using drift diffusion models, as described in the General Methods. We hypothesized that participants would recognize identities encoded through multiple expressions better than those with a single exemplar. Moreover, we expected this memory advantage to be influenced by the specific emotions presented, either in terms of threat-signaling relevance (i.e., anger and fear) or stimulus arousal.

General methods

Participants

Young healthy adults were recruited from the McGill Psychology extra-credit participant pool and from a McGill University community (students and alumni) Facebook group. Participants completed the studies online on a JATOS server (Lange et al., 2015), hosted by the International Laboratory for Brain, Music, and Sound Research (BRAMS), using the jsPsych library (de Leeuw, 2015). They were all fluent in English, had normal hearing and (corrected-to-) normal vision, reported no prior diagnosis or treatment of psychiatric/neurological conditions, and received course credits or monetary compensation (CAD10). Each subject took part in only one of the studies. The studies were approved by the Faculty of Medicine Research Ethics Office at McGill University.

Stimuli

Vocal stimuli were audio-only recordings from the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) (Livingstone & Russo, 2018). The recordings consisted of 24 speakers (12 female, aged 21–33) uttering a semantically neutral sentence ("Kids are talking by the door"), in four emotional (fear, happiness, sadness, and anger, the strongly expressive version) and neutral expressions (i.e., prosody). Accent variability was already minimized given that the database was designed and created with a neutral North American accent, to avoid participants using accent as a voice recognition shortcut (Gluszek & Dovidio, 2010). Speech clips were trimmed to exclude silence in the beginning and end of the original recordings in Praat v6.1.04 (Boersma & Weenink, 2019). Loudness was then normalized using the Loudness Toolbox (Genesis S.A.) in Matlab 2017b. The final speech clips had a mean duration of 1.75 s (SD = 0.30 s; range: 1.26–2.91 s).

Face images were taken from the Karolinska Directed Emotional Faces (KDEF) database (Lundqvist et al., 1998).

Table 1 Overview of the emotional expression combinations used in the studies

Study	Encoding		Recognition
	Uni	Multi	
Main Study (N = 547)	angry / fearful / happy / sad *(between subjects)*	angry, fearful, happy, sad	neutral
Follow-up Study 1 (N = 172)	1.1 fearful, sad	-	neutral
	1.2 fearful, neutral	-	happy
Follow-up Study 2 (N = 174)	angry / fearful *(between subjects)*	angry, fearful, happy	neutral

Forty-eight actors (24 female, aged 20–30) were selected based on highest emotion categorization accuracy on the four emotional expressions (Goeleven et al., 2008), taken into consideration together with the criteria of as little presence of significant features/marks on images that enabled easy recognition as possible (for the list of selected actors and additional information of the stimuli, see Supplementary Information [SI]). The exterior of faces (e.g., shoulder) was removed using Adobe Photoshop CS5.1 (Adobe Systems, San Jose CA), to achieve a uniform face size, contrast and resolution (see Sergerie et al., 2005, 2006).

Normative values for emotional judgments on the face and voice stimuli, obtained from three validation studies (Goeleven et al., 2008; Sutton et al., 2019 for faces; Livingstone & Russo, 2018 for voices), are provided in SI Table 1.

Procedure

Participants were instructed to complete the online studies on a computer browser in a quiet environment. There was no interaction between participants and experimenters during the procedure. The experiment was designed to work in the most common browsers and operating systems. Access to a keyboard and headphones was required for participation.

In all three studies, each participant completed two single-modality modules (one face, one voice) in full-screen mode sequentially, with the order counterbalanced. Each module contained an encoding and a recognition phase. Each phase was programmed to automatically start three seconds after the completion of the preceding one. While there were self-paced text instructions at beginning of each phase, participants were told not to take long pauses on the instruction pages. Post-hoc tests also confirmed that the time spent on instructions between encoding and recognition phases did not have a significant influence on performances. In the face module, participants were presented a series of emotional face images from a number of identities (half female), each for 1.5 s, and were asked to judge the age bracket of each face (younger or older than 30 years) by pressing a key. Immediately after the encoding phase, an old/new recognition test took place, where a novel-expression face image of previously encoded identities and an equal amount of novel identities was presented in pseudo-random order. Additionally, subjects were asked to give a binary confidence rating (*Sure/Unsure*) on each response they made during recognition. In both phases, the following trial started 0.5 s after a response was recorded. The structure of the voice module was identical to the face module. Participants were instructed beforehand that there was no relation between the stimuli of both modules. A schematic of the experimental trials is shown in Fig. 1. Study-specific manipulations on encoding conditions are further described in detail under each study and summarized in Table 1.

Throughout the online experiment, the assignment of response key in both the encoding (age) and recognition trials was randomized within subject (e.g., an 'Older than 30' or 'Old' response may be linked with 'J' key in one trial, but with 'K' key in another). This was to avoid potential subject-specific response lateralization tendency (e.g., tendency to press a given key). However, the key assignment for the confidence rating was fixed within subject and randomized across subjects. Lastly, a short practice session was implemented before the actual experiment, containing the encoding and recognition phases of both modality modules, with fewer and entirely different stimuli (for details of the practice session, see Practice Session section in SI). Feedback on recognition accuracy was given at the end of each practice trial to ensure participants understood the task and were comfortable with the stimuli.

Dependent measures

Response choices and response times (RTs) for both encoding and recognition tasks, as well as the binary confidence ratings during the recognition test, were recorded using customized jsPsych plugins for subsequent analysis. Data used for the following analyses are available from the authors upon request.

Data analysis

Recognition accuracy

Binary measures of accuracy (correct/incorrect) and confidence (Sure/Unsure) during the recognition test were analyzed on a trial-by-trial level, using a generalized linear mixed model (GLMM) with a logit link function. Fixed-effect factors were defined according to the experimental design, as described in the study-specific Methods sections below. The random effect structure included random intercepts of subject and stimulus identity, with the latter accounting for possible differences in identity-specific distinctiveness and to allow for the generalization of findings beyond the material used here (Clark, 1973; Baayen et al., 2008). The analysis was implemented in R with the *lme4* R package (Bates et al., 2015). The omnibus effects of modelled fixed factors were obtained via *Anova* function (type III) from the *car* package (Fox & Weisberg, 2019).

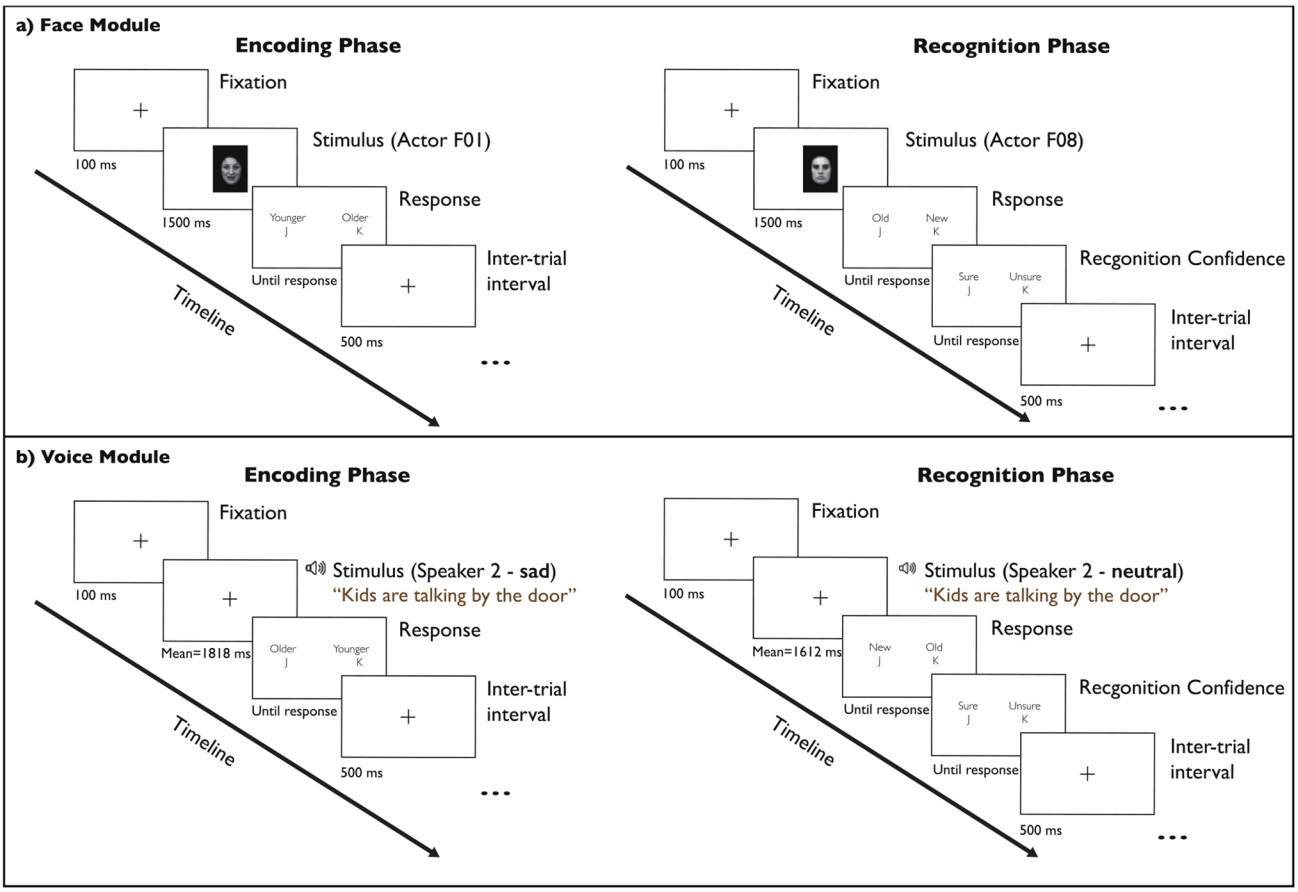

Fig. 1 Procedure and example trial of the experimental task in the studies. (a) and (b) depict the example trials for both encoding and recognition phases, in face and voice modules respectively

Drift diffusion models (DDMs)

We employed DDMs (Ratcliff & McKoon, 2008) to analyze subjects' binary choices, combining their trial-specific responses and RTs, as a process in which evidence is accumulated in each trial towards one of the two possible responses, until a threshold is reached, and a decision is made. A key parameter of this model is the drift rate (v), defined as the rate of information accumulation, determined by the quality of the extracted stimulus information. DDMs have been extensively used to examine cognitive processes such as memory and decision making (for a review, see Ratcliff et al., 2016) and, more recently in emotion research (e.g., Williams et al., 2023; Mueller & Kuchinke, 2016, for emotional face and word perception, respectively). In the context of recognition memory, drift rate can be interpreted as the quality of match between the tested stimulus and memory, which is usually expected to vary among different experimental conditions. The other main parameters of the model are the starting point (z), usually used to model the changes of response proportions, and the boundaries (a), which are assumed to vary in cases of an instruction change (e.g., emphasis on response speed or accuracy).

We adapted a recent variation of diffusion models, D*M (Verdonck & Tuerlinckx, 2016), using the DstarM package in R. Specifically, we ran D*Ms on trial-level response and RT data for each module of each participant separately, to estimate the parameters v, a, z, and within-subject variability of v (sv) and z (sz). Drift rates were modeled to vary among recognition conditions, while the other parameters were held constant as they were not hypothesized to change across experimental conditions. The estimated subject-condition-specific drift rates were then submitted to linear mixed models (LMMs), with fixed factors including modality and other study-specific variables. The random effect structure contained a random intercept of subject. Results of the LMM was obtained via *anova* function (type III) from *lmerTest* package (Kuznetsova et al., 2017).

For both LMM and the GLMM results, whenever significant interactions were present, post-hoc pairwise tests (for categorical variables) or post-hoc tests of linear trends (for continuous variables) were conducted using the *emmeans* R package (Lenth, 2020) (Bonferroni-Holms correction

implemented when needed). Effect sizes were obtained with the *RESI* package (Vandekar et al., 2020) for fixed omnibus effects in the models (denoted as S index), and *emmeans* package for post-hoc pairwise tests on interactions: odds ratio [OR] for post-hoc tests in GLMMs, and Cohen's *d* in LMMs.

Stimulus-based physical feature analysis

In order to investigate the possibility that (part of) the results could be explained by intrinsic stimulus physical characteristics, we conducted a stimulus-based similarity analysis for voice and face stimuli separately, as a function of emotional expression. For voices, we extracted 17 acoustic parameters from each speech clip, including stimulus duration and multiple formant frequency descriptive statistics (see Xu & Armony, 2021 for details) using Praat v6.1.04 (Boersma & Weenink, 2019). In the case of faces, we conducted a pixel-based principal component analysis on the 240 face images for dimension reduction. We kept 56 principal components ("eigenfaces"), explaining 90% of the total variance, with the corresponding coefficients per stimulus characterizing their physical attributes. Each stimulus was therefore represented in a multidimensional (17 for voices, 56 for faces) feature space (Armony et al., 2007; Baumann & Belin, 2010; Latinus et al., 2013; Sergerie et al., 2005).

First, we examined whether stimuli within certain emotions were more similar among each other than other emotions, by calculating the average cosine similarity between each stimulus and all the other same-emotion stimuli and. Those values were then analyzed in a one-factor (emotion) LMM, with a random identity intercept. Next, we examined whether each identity's physical change from a given emotion to its neutral counterpart differed among emotions. To do so, we computed the cosine similarity between each emotional (angry, fearful, happy, and sad) and neutral expression, both within and between identities. Then, the difference of within- and between-identity similarity per identity per emotion, representing an identity's distinctiveness from other identities in one emotion, were entered in a one-factor (emotion) LMM, with a random identity intercept.

Main study

Methods

Participants

A cohort of 556 healthy individuals (523 and 33 from the Participant pool and Facebook post, respectively; see General Methods) took part in the study. One participant was excluded because of repeated participation. Eight participants were excluded due to incomplete participation. Thus, the final sample consisted of 547 participants (481 female aged 18–31 years: Mean [M] = 20.5, Standard Deviation [SD] = 1.5; 66 male aged 18–32 years: M = 21.1, SD = 2.3).

Procedure

In the face module, participants first viewed 96 face images from 24 randomly selected identities (half female); that is, during this encoding phase, each identity was presented four times, intermixed with presentations of others. Twelve identities were randomly assigned to the *Multi* (exemplar variability) condition, and thus presented in four distinct emotional expressions (fearful, happy, angry and sad) once each. The other 12, assigned to the *Uni* condition, were presented with a single emotional expression (i.e., same stimulus) four times. The single emotion used in this condition was pseudo-randomized across participants, resulting in four groups of participants (i.e., *Uni*-Fearful [$N = 145$], *Uni*-Happy [$N = 144$], *Uni*-Sad [$N = 129$], *Uni*-Angry groups [$N = 129$]). The *Uni* and *Multi* trials were pseudo-randomly intermixed, and the distribution of lags (i.e., number of intervening stimuli) between same-identity exemplars did not differ between the *Uni* and *Multi* conditions. Moreover, both presentation order of emotions and first-order transition of same-identity emotional stimuli, were balanced across participants among *Multi* trials. The old/new recognition test consisted of the 24 encoded identities and 24 novel ones, all with a neutral expression. The voice module shared the identical structure but with 12 identities presented during encoding, and 24 (half new) during recognition. We used a smaller number of identities to ensure adequate memory performance, as determined by pilot experiments and previous studies (Xu & Armony, 2021). The assignment of identities to each category (*Uni*, *Multi* or *New*) was random and counterbalanced across participants.

Data analysis

Encoding Response times (RTs) from the encoding task (age judgment) were analyzed to test whether there were stimulus- and/or identity-specific implicit memory (priming) effects. RTs were cleaned by excluding trials with RTs that were 3 SD beyond mean value per subject. Then, a linear regression on same-identity RTs was performed for each encoded identity per subject (Lorch & Myers, 1990). The regression coefficients (RT-slopes) were then submitted to an LMM, with encoding variability (*Multi*/*Uni*) and modal-

ity (visual/auditory) as fixed within-subjects factors, and *Uni*-emotion as fixed between-subjects factor.

We also computed within-identity consistency in the age judgment responses across the 4 presentations, the Age Consistency Score (ACS), as follows: ACS = 1-[#presentations - max(#Old, #Young)]. That is, the possible values were 1 (all 4 age responses were the same), 0 (3 out of 4 responses were the same), and − 1 (2 out of 4 responses were the same).

The subject- and identity-specific RT-slopes and ACSs were then entered as covariates in the recognition accuracy model, described below, to determine whether priming and/or age consistency predicted subsequent identity recognition memory and, if so, whether this effect interacted with the encoding condition (*Multi*/*Uni*) .

Recognition of old identities Overall recognition accuracy (across both new- and old-identity trials) was analyzed in a simple mixed ANOVA with modality as a within-subjects factor and *Uni*-emotion group as a between-subjects factor. Three (G)LMM models were constructed within old identity trials, namely on recognition accuracy, DDM-derived drift rates, and recognition confidence.

(1) Recognition accuracy.

In this GLMM, fixed effect structures consisted of two within-subject factors, encoding exemplar variability condition (*Uni/Multi*) and modality (visual/auditory), and one between-subjects factor, *Uni*-emotion group (*Uni*-Angry/*Uni*-Fearful/*Uni*-Happy/*Uni*-Sad). The two encoding indices representing identity priming and age consistency (RT-slope and ACS) were entered into the model as covariates.

To ensure that any difference (or lack thereof) in memory accuracy between *Uni* and *Multi* conditions was truly reflective of the entire dataset (as opposed to the possibility of being driven by a small number of extreme or high-leverage subjects), we performed a series of n-jackknife resampling cross-validations: for each participant group, 1000 random subsets of n subjects of the original sample were selected ($n = 5$–95% of the original sample, in steps of 5%). At each data fraction, the mean jackknife estimate of the *Multi-Uni* accuracy difference and its 95% confidence interval were calculated.

(2) Drift rates.

Subject- and condition-specific drift rates obtained from the drift diffusion model were entered into an LMM with condition and modality as within-subject factors and *Uni*-emotion as a between-subject one.

(3) Recognition confidence.

Recognition confidence was analyzed in a GLMM with encoding variability condition, modality and accuracy as within-subjects, and *Uni*-emotion as between-subjects, factors. In order to validate the hypothesized relationship between confidence rating and response times (Shaw, McClure, & Wilkens, 2001; Robinson et al., 1997; Weidemann & Kahana, 2016), we added log-transformed RTs as covariate in the model.

Detection of new identities For new identity trials, (G) LMMs with a similar structure were constructed on the same three measures: recognition accuracy, DDM-derived drift rates, and recognition confidence. Importantly, as new identity trials did not have the encoding exemplar variability condition, the models only contained modality and *Uni*-emotion as within- and between-subjects fixed factors, respectively. As with old identities, we included response accuracy and log-RT as covariates in the recognition confidence GLMM.

Stimulus-based arousal-rating analysis Finally, to assess the potential impact of stimulus arousal on identity memory, we performed linear models on recognition accuracy of *Uni* identities, using the arousal ratings of stimuli from two prior validation studies (Sutton et al., 2019 for faces; Livingstone & Russo, 2018 for voices). To account for the discrepancies between rating ranges of the two datasets (1–5 points for voices, and 1–9 points for faces), the rating data were normalized within each modality prior to modelling. Linear models were then performed on subject-averaged recognition accuracy with arousal rating, emotion, and modality as fixed factors.

Results

Encoding: implicit memory (priming)

The LMM on estimated identity-repetition RT slopes (an index of priming; see Methods for more details) as a function of modality (visual/auditory), encoding variability (*Multi*/*Uni*) and *Uni*-emotion (participant groups) revealed that in all cases the slopes were significantly negative (*Uni*: b = -0.067, SE = 0.004, z = -17.60, $p < .001$, Cohen's $d = 0.26$; *Multi*: b = -0.052, SE = 0.004, z = -13.59, $p < .001$, Cohen's $d = 0.20$), reflecting identity-specific priming (i.e., faster responses as a function of repeated presentation of the same individual). Nonetheless, a main effect of encoding variability ($F[1,19133] = 15.63$, $p < .001$, S = 0.31) indicated that, as expected, this effect was stronger in the *Uni* than *Multi* condition. Additionally, there was a main effect of modality ($F[1,19133] = 139.85$, $p < .001$, S = 0.59), due to the steeper slopes for voices than faces.

Recognition: identity memory

Overall accuracy A 2 (modality) by 4 (*Uni*-emotion) mixed ANOVA on overall accuracy revealed a significant modality effect (F[1,543] = 739.90, p < .001, $\eta_p^2 = 0.58$), due to a better memory for faces than voices, with no *Uni*-emotion main effect (F[3,543] = 0.65, p = .58, η_p^2= 0.004) or interaction (F[3,543] = 0.19, p = .90, η_p^2= 0.001). Planned t-tests confirmed that overall memory accuracy was significantly above chance level for both modalities (voice: t[546] = 130.21, 95% CI = [0.53, 0.55], p < .001; face: t[546] = 172.28, 95% CI = [0.69, 0.70], p < .001; descriptive stats in SI Table 2).

Recognition of old identities Results of the GLMM (see SI Table 3 for full results) on recognition accuracy with encoding variability, modality as within-subject, and *Uni*-emotion as between-subject factors, as well as encoding RT-slope and Age Consistency Score as covariates, revealed significant main effects of modality (χ^2[1] = 7.54, p = .006, S = 0.11; better memory for faces than voices), encoding exemplar variability (χ^2[1] = 6.60, p = .010, S = 0.10; better memory for *Multi* than *Uni* identities), RT-slope (χ^2[1] = 8.51, p = .004, S = 0.12; stronger identity priming predicted better recognition), and ACS (χ^2[1] = 23.17, p < .001, S = 0.20; higher intra-identity consistency in age judgment predicted higher recognition accuracy). Importantly, the exemplar variability effect was qualified by a significant variability-by-emotion interaction, χ^2(3) = 11.25, p = .010, S = 0.12. Post-hoc tests on encoding condition (see SI Table 4, and Fig. 2a for visualization) indicated that this interaction was due to the memory advantage for *Multi* identities in all groups (p's ≤ 0.003, OR's ≥ 1.29) except the *Uni*-Sad group (b = -0.003, SE = 0.07, z = 0.05, p = .96, OR = 1.00).

To rule out the possibility that the difference between *Uni*-Sad and the other groups could have been driven by subject-specific confounding effects (e.g., extreme or high-leverage data points), we performed an n-jackknife supplementary analysis. This analysis showed that the encoding variability effect (or lack thereof) was stable even for small random subsets of the original sample (Fig. 3). Specifically, for the *Uni*-Angry, Fearful and Happy groups, the *Multi* advantage remained significant with only about 50% or less of the original sample sizes. In contrast, for the *Uni*-Sad group, the 95% CIs included zero for all subsamples, indicating a reliable lack of accuracy difference between encoding conditions.

Subject- and condition-specific drift rate estimates from the diffusion models (D*Ms) are summarized in Fig. 2b (descriptive stats in SI Table 2). The LMM for old identities revealed both main effects of encoding condition (F[1,1581.1] = 39.03, p < .001, S = 0.26) and modality (F[1,1583.5] = 300.94, p < .001, S = 0.74), as well as emotion-by-variability (F[3, 1580.9] = 4.31, p = .005, S = 0.13) and modality-by-condition (F[1, 1579.6] = 6.95, p = .008, S = 0.10) interactions (SI Table 3). Consistent with the accuracy results, the emotion-by-variability interaction was due to a significantly larger drift rate in the *Multi* than *Uni* condition, in all (p's < 0.01, Cohen's d > 0.2) but the *Uni*-Sad group (b = 0.05, SE = 0.072, t[1603] = 0.76, p = .45, Cohen's d = 0.068) via post-hoc pairwise tests (SI Table 4). The modality-by-variability interaction was due to the fact that although both modalities showed a larger drift rate for *Multi* than *Uni* identities, the effect was larger for faces (voice: b = 0.13, SE = 0.05, t[1595] = 2.55, Cohen's d = 0.16; face: b = 0.31, SE = 0.05, t[1594] = 6.29, Cohen's d = 0.39).

The GLMM on response confidence (*Sure*/*Unsure*) yielded a significant effect of accuracy (χ^2[1] = 21.79, p < .001, S = 0.19), modality (χ^2[1] = 11.06, p = .001, S = 0.14), and log-transformed RTs (χ^2[1] = 722.20, p < .001, S = 1.15), as well as variability-by-modality (χ^2[1] = 4.62, p = .032, S = 0.081), modality-by-accuracy (χ^2[1] = 11.51, p < .001, S = 0.14), and variability-by-modality-by-accuracy (χ^2[1] = 6.08, p = .014, S = 0.096) interactions (see SI Table 5). The significant effect of RTs indicated faster responses for *Sure* trials, supporting the hypothesis that they can serve as a proxy for response confidence (Shaw, McClure, & Wilkens, 2001; Robinson et al., 1997; Weidemann & Kahana, 2016). Post-hoc tests on the three-way interaction indicated that participants were significantly more confident when successfully recognizing *Multi*-encoded than *Uni*-encoded identities for faces (b = 0.13, SE = 0.06, z = 2.35, p = .019, OR = 1.14), but not for voices (b = 0.06, SE = 0.08 z = 0.82, p = .41, OR = 1.06). No difference in incorrect response confidence between *Multi* and *Uni* identities was found in either modality (voice: b = 0.001, SE = 0.09, z = 0.01, p = .99, OR = 1.00; face: b = -0.09, SE = 0.08, z = -1.22, p = .22, OR = 0.91).

Detection of new identities Similar analyses were conducted for performance on new identities, but with only two fixed factors, modality and *Uni*-emotion group, as these identities were not related to the expression-variability manipulation in the encoding phase. The accuracy GLMM (full stats in SI Table 3) revealed a significant effect of modality (χ^2[1] = 37.12, p < .001, S = 0.28; higher accuracy for faces than voices), but no main effects interactions involving *Uni*-emotion groups (p's > 0.5, S < 0.001).

The drift-rate LMM (full stats in SI Table 3) also yielded a modality effect (F[3,535.4] = 533.38, p < .001, S = 0.98): whereas the drift rates for new faces were significantly

negative (i.e., towards a "New" response; b = -0.81, SE = 0.03, t[1061] = -27.18, p < .001, Cohen's d = 1.25), those for voices were positive (b = 0.10, SE = 0.03, t[1060] = 3.43, p < .001, Cohen's d = 0.16), reflecting a tendency to respond "Old" towards new voices (Fig. 2b).

In the case of the binary confidence responses, the GLMM yielded main effects of accuracy (χ^2 [1] = 11.56, p < .001, S = 0.14) and modality (χ^2 [1] = 67.14, p < .001, S = 0.35), as well as an interaction between the two (χ^2 [1] = 39.63, p < .001, S = 0.27). Post-hoc tests on the interaction showed that whereas correct responses of faces were made with higher confidence than incorrect ones (b = 0.57, SE = 0.05, z = 12.35, p < .001, OR = 1.77), the opposite was true for voices (b = -0.45, SE = 0.06, z = -7.10, p < .001, OR = 0.64). There was also a significant effect of the log-RT covariate (χ^2 [1] = 742.64, p < .001, S = 1.16; see SI Table 5), confirming the relation between implicit (faster RTs) and explicit (Sure/Unsure choice) confidence measures.

Stimulus-based physical features A one-way LMM on the within-emotion similarity for vocal stimuli showed a significant effect of emotional prosody (F[4,92] = 9.24,

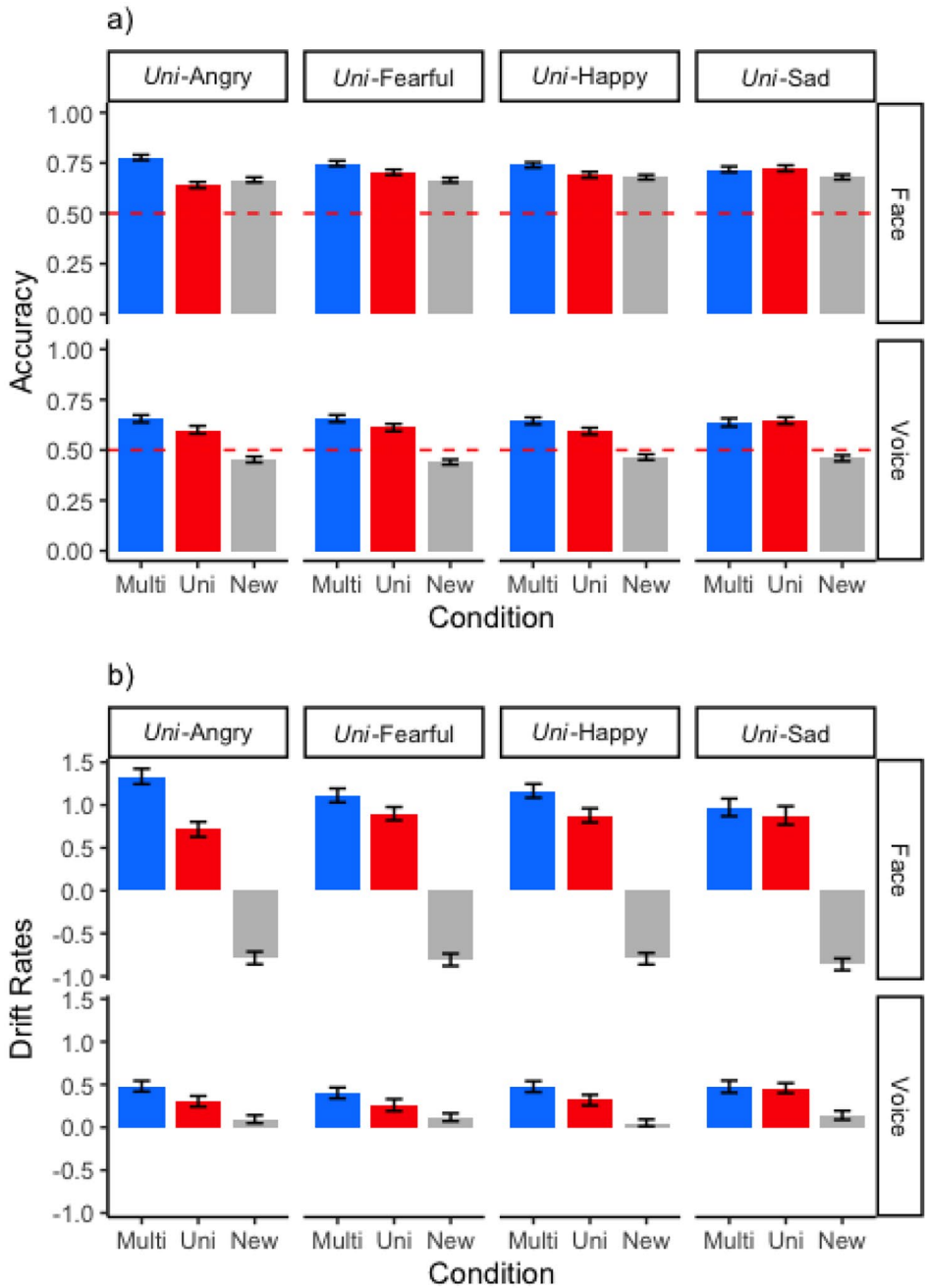

Fig. 2 Averaged recognition accuracy (*a*) and DDM-derived drift rates (*b*) by encoding variability in each *Uni*-emotion group and modality (dashed horizontal line in *a* indicated chance-level accuracy)

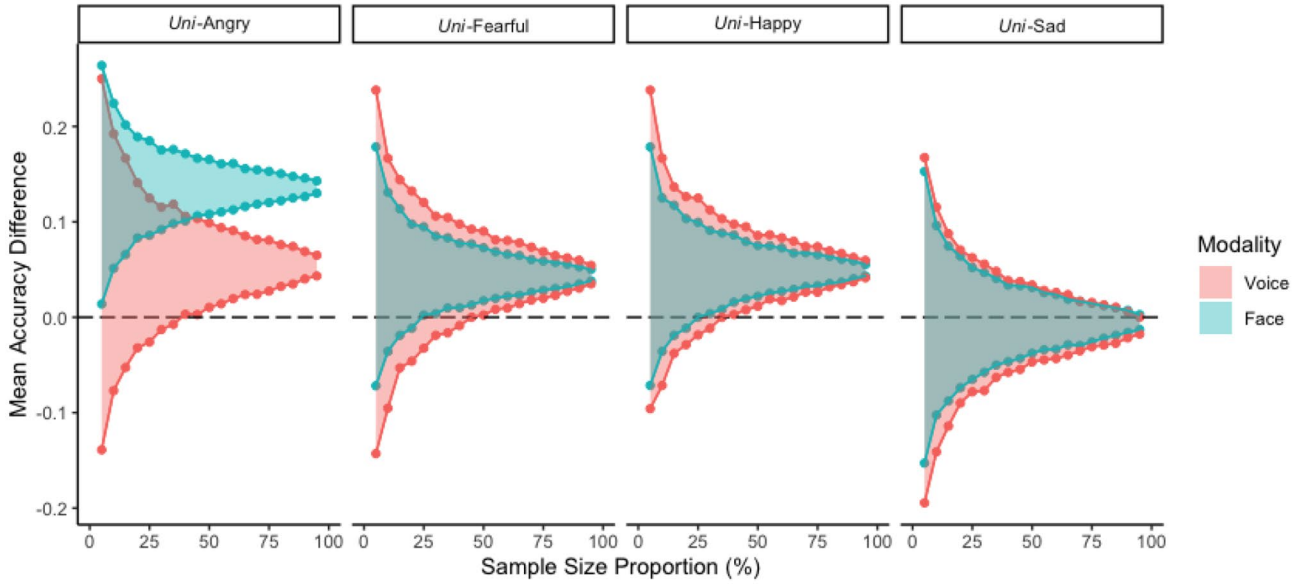

Fig. 3 Confidence intervals (95%) of accuracy difference between *Multi* and *Uni* conditions from the Main study, using an n-jackknife subsampling approach with 5% step-wise sample sizes from 5–95%

$p < .001$, S = 0.52) (descriptive statistics of computed cosine similarity in SI Table 6). Post-hoc pairwise *t*-tests revealed that this was driven by within-neutral similarity being significantly larger than within-sad, fearful and angry similarities ($t[92]$'s > 3.5, p's < 0.003, Cohen's d's > 1.1) as well as within-happy similarity being larger than within-fearful similarity ($t[92] = 3.06$, $p = .029$, Cohen's $d = 0.88$; complete pairwise test statistics listed in SI Table 7). A one-way LMM on the difference of within- and between-identity similarity between emotional and neutral stimuli did not reveal a significant effect of prosody ($F[3,69] = 1.12$, $p = .35$, S = 0.054), suggesting that the speaker distinctiveness in acoustic similarity between emotional and neutral speech did not differ among the four emotions.

The same analyses on face-image derived principal components (see SI Table 6 for cosine-similarity descriptive statistics) also showed a significant effect of emotional expression on within-emotion average similarity ($F[4,188] = 2.52$, $p = .043$, S = 0.16), which was driven by a significantly higher within-happy than within-neutral face similarity ($t[188] = 3.04$, $p = .027$, Cohen's $d = 0.62$; see complete pairwise test statistics in SI Table 7). For the emotional-to-neutral identity distinctiveness, the LMM did not reveal a significant emotion effect ($F[3,141] = 0.20$, $p = .89$, S < 0.001). Taken together, these results support the notion that the behavioral recognition patterns reported above were unlikely to be due to intrinsic physical differences of the stimuli in either modality.

Stimulus-based arousal rating The linear model with stimulus-level accuracy of all *Uni* recognition trials as dependent variable and emotion, modality and stimulus arousal intensity rating as factors, yielded a main effect of modality ($F[1,269] = 17.73$, $p < .001$, S = 0.22; higher accuracy for faces than voices) and arousal rating ($F(1,269] = 22.15$, $p < .001$, S = 0.27; descriptive statistics and fixed factor effects of the model are shown in in SI Tables 1 and 8, respectively). The latter effect reflected a negative correlation between stimulus emotional arousal and identity recognition memory for both faces and voices, independent of the specific emotion expressed, as shown in Fig. 4.

Summary

Results from the Main Study revealed a significant advantage of emotional expression variability (*Multi* condition) during encoding for the subsequent recognition of facial and vocal identities, relative to the repeated presentation of angry, happy and fearful, but not sad expressions (*Uni* condition). In the *Uni*-Sad group, there was no significant difference in recognition accuracy between *Multi* and *Uni* encoding conditions. As the *Multi* condition was the same for all *Uni* emotion groups, we speculated that the absence of an advantage for this condition in the *Uni*-Sad group was driven by whatever feature distinguishes this emotion from the others (happiness, anger and fear), one likely candidate being emotional intensity or arousal (e.g., Kensinger, 2009). This possibility was supported by the significant

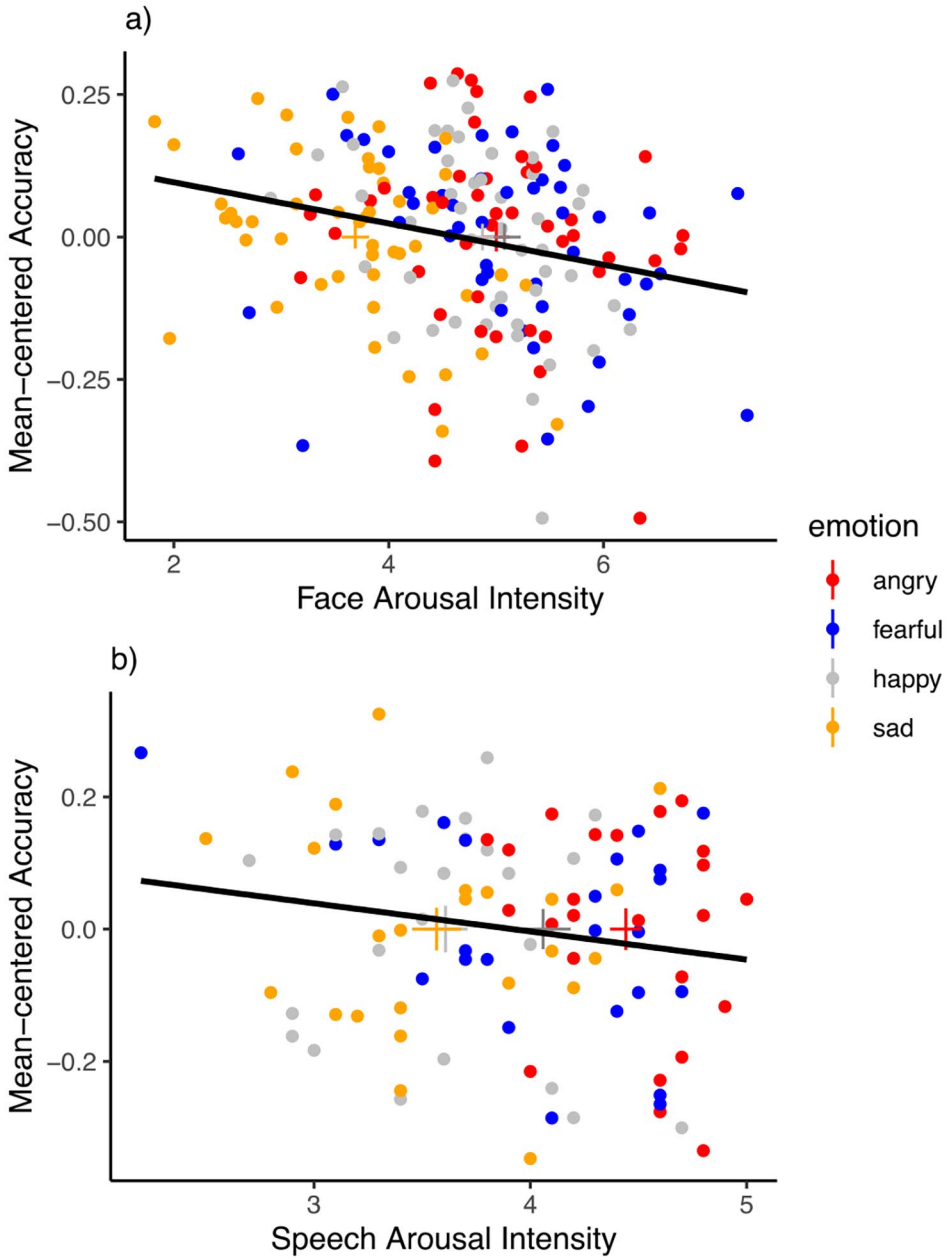

Fig. 4 Scatterplot and estimated regression line of stimulus arousal intensity ratings against across-subject mean accuracy after accounting for accuracy difference among *Uni*-emotion groups (crosses and the arm lengths represent mean and standard error of intensity ratings per stimulus emotion category)

negative correlation between stimulus arousal, or emotional intensity, and identity recognition accuracy across expressions and modalities. That is, individuals whose emotional expressions were judged to be less arousing were better recognized later, when presented in a novel, neutral expression, than those considered to express high intensity emotions. However, the between-group nature of our design did not allow for directly testing this possibility; we therefore conducted two additional studies, each consisting of two related experiments, focusing on specific combinations of emotional expressions based on their arousal level (Table 1).

Follow-up study 1: high vs. low arousal emotions

I: fearful vs. sad

In this experiment, we directly compared recognition performance of identities presented with only one expression during encoding, either sadness or fear, in a within-subject design. Based on the results from the Main Study, we predicted a memory recognition advantage for sad-encoded identities, compared to those encoded with a fearful expression.

Methods

Eighty-six (18–26 years: M = 20.5, SD = 1.6; 74 female) new participants were recruited through the McGill Psychology extra-credit participant pool. The overall procedure was similar to the original experiment in the *Uni*-Sad group, consisting of both face and voice separate modules, but with the original *Multi* encoding condition replaced by a *Uni* condition in fearful expressions. Hence, all encoded identities (same amount as in the Main Study) were presented four times, half with a single sad exemplar and the other with a fearful one. To test our hypothesis, we focused our analysis on the trial-level recognition response and DDM-derived drift rates of old identity trials. The modeling structure was similar as in the Main Study, but with only modality, and the encoding emotion (fearful vs. sad) as within-subject factors.

Results

The GLMM on recognition accuracy yielded main effects of modality (χ^2 [1] = 10.15, p = .001, S = 0.33) and encoding emotion (χ^2 [1] = 49.05, p = .003, S = 0.31) without a significant interaction (χ^2 [1] = 0.63, p = .43, S < 0.001), driven by an overall better recognition for faces, and for neutral- than fearful-encoded identities (Fig. 5). Likewise, the LMM on DDM-derived drift rates revealed significant main effects of modality (F[1,251.37] = 62.45, p < .001, S = 0.84; larger drift rates for faces than voices) and encoding emotion (F[1,251.36] = 8.62, p = .004, S = 0.30; larger drift rates for neutral- than fearful-encoded identities), without a significant interaction (F[1,251.37] = 0.82, p = .37, S < 0.001).

Results

The GLMM on recognition accuracy yielded main effects of modality (χ^2 [1] = 8.99, p = .003, S = 0.30) and encoding emotion (χ^2 [1] = 8.34, p = .004, S = 0.29) without a significant interaction (χ^2 [1] < 0.001, p = .99, S < 0.001), suggesting an overall better recognition for face, and a better recognition for sad than fearful encoded identities (Fig. 5a). For subject-level condition-specific drift rates, the LMM revealed significant effects of both modality (F[1,252.61] = 50.46, p < .001, S = 0.76) and encoding emotion (F[1,251.31] = 6.35, p = .012, S = 0.25), without a significant interaction (F[1,251.29] = 0.28, p = .60, S < 0.001), due to larger drift rates for faces than voices, and for sad- than fearful-encoded identities (Fig. 5b; descriptive stats in SI Table 2).

II: fearful vs. neutral

Methods

To further confirm, and extend, the results from the previous experiment, we conducted a similar experiment but using neutral expressions, instead of sad, as the low-arousal expression. To keep the emotional expressions different between encoding and recognition, as in the previous experiments, we used happy faces/voices in the recognition test. The recruitment process remained the same, with a group of 86 new participants completing the experiment (18–29 years: M = 20.3, SD = 1.7; 76 female). Analysis on recognition response and drift rates was the same as in the Fearful vs. Sad experiment.

Follow-up study 2: high-arousal *Multi* vs. *Uni*

Results from the Main Study and Follow-up Study 1 point to a key role of arousal in identity recognition. Because the *Multi* condition in the Main Study included both high- (anger, fear and happiness) and low- (sadness) arousal emotions, there remains the question of whether the better performance for this condition, compared to high-arousal *Uni* ones, was due to the presence of the low arousal expression in the former. Thus, to directly test this possibility, we conducted an additional online experiment with the same paradigm and recruitment criteria, without using any sad stimuli. Specifically, each identity was presented three, instead of four, times, during encoding. In the *Multi* condition, three emotional exemplars (happiness, fear, and anger) of each identity were presented once; in the *Uni* condition, either one angry or fearful exemplar for each identity was presented three times, which served as a between-subjects variable (i.e., *Uni*-Fearful vs. *Uni*-Angry). We expected the multiple exemplar advantage to be present for both groups.

Methods

A total of 174 new participants (18–31 years: M = 20.5, SD = 1.7; 154 female) were recruited via the McGill Psychology extra-credit participant pool under the same criteria as previous studies. Half of the participants (N = 87) were presented with angry exemplars in the *Uni* condition, while the other half with fearful exemplars. We analyzed recognition accuracy and DDM-derived drift rates of old identity trials in the same manner as before.

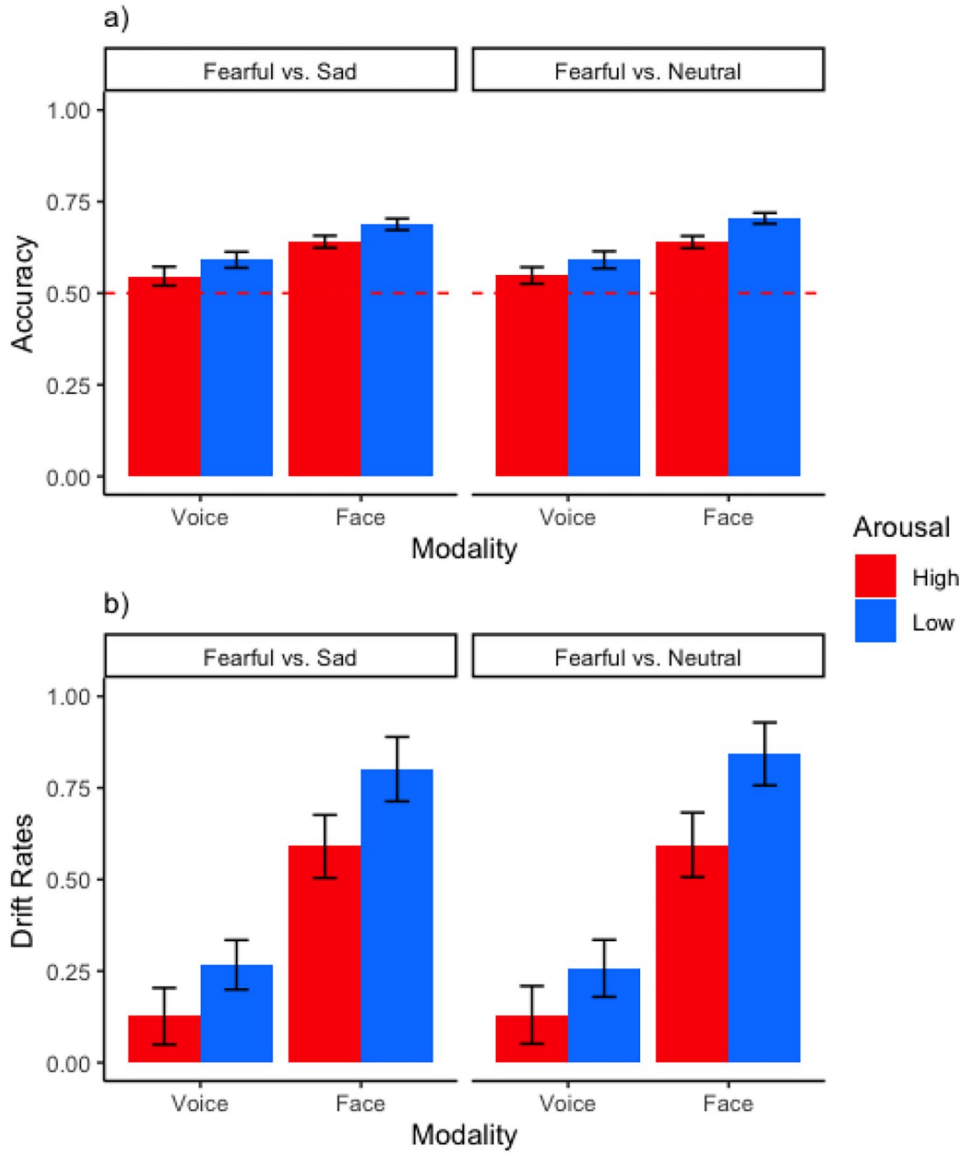

Fig. 5 Averaged recognition accuracy (*a*) and DDM-derived drift rates (*b*) by encoding emotion arousal level per modality from Follow-up Study 1 (dashed horizontal line in *a* indicated chance-level accuracy)

Results

The GLMM on recognition accuracy, with modality and exemplar variability as within-subject and *Uni*-emotion as between-subjects factors (SI Table 9), revealed significant modality ($\chi^2[1] = 8.46$, $p = .004$, S = 0.21) and variability ($\chi^2[1] = 20.28$, $p < .001$, S = 0.33) effects, driven by, as in the case of the main study, a more accurate recognition for faces, and for *Multi*-encoded identities (Fig. 6a). The LMM on drift rates yielded a similar pattern, with significant effects of modality ($F[1,503.41] = 127.12$, $p < .001$, S = 0.33) and variability ($F[1, 502.05] = 20.42$, $p < .001$, S = 0.85). Additionally, there was a significant *Uni*-emotion effect ($F[1, 167.14] = 4.24$, $p = .041$, S = 0.14), due to overall larger drift rates in the *Uni*-Fearful than *Uni*-Angry groups, as illustrated in Fig. 6b.

General discussion

Implicit and explicit identity memory

As expected, we observed significant priming in repeated presentations of the same face and voice expression, as indexed by negative RT slopes of the age judgment task. Notably, a similar effect, albeit weaker, was observed when the same individual was presented with different emotional expressions. That is, over the course of the encoding session, participants developed expression-independent implicit memory for previously-encountered individuals.

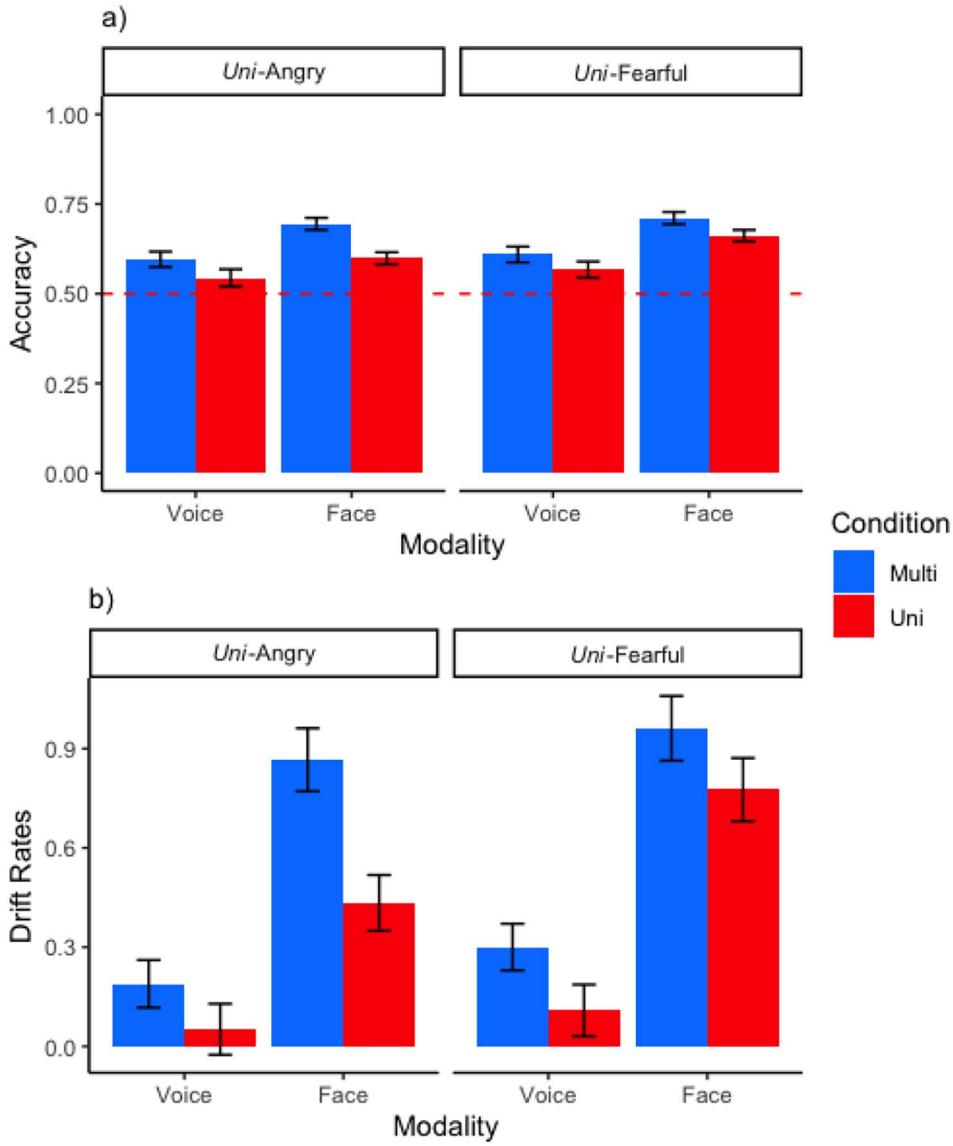

Fig. 6 Averaged recognition accuracy (*a*) and DDM-derived drift rates (*b*) by encoding variability per modality from Follow-up Study 2 (dashed horizontal line in *a* indicated chance-level accuracy)

Interestingly, the magnitude of the priming effect during encoding positively predicted subsequent recognition memory for both faces and voices. These findings contribute to a literature that has yielded conflicting results (e.g., Li & Jiang, 2020; Miyoshi et al., 2014), providing new and strong support for a direct relation between implicit and explicit memory processes (Turk-Browne et al., 2006; Gagnepain et al., 2008). Further evidence comes from the task itself: age judgment consistency predicted recognition accuracy. That is, when a participant judged the same individual more consistently as being younger or older than 30 years of age, regardless of whether appeared with the same or different emotional expression, the more likely they were to remember that individual later. As the stimulus identities were presented with a never-encountered neutral expression in the recognition test, their successful recognition required the formation of a stable expression-independent representation of that individual. Consistently assigning an identity to a given age bracket during repeated encounters, either with the same or different expression, could be taken as evidence, albeit indirect, that such an individual-specific representation was being formed.

Most of the effects discussed here and in the next sections were present in both face and voice modules. Nonetheless, some modality differences were observed. In general, accuracy was higher for faces than voices; this is not surprising given that voice is considered a weaker identity cue and voice recognition is more error-prone (e.g., Stevenage & Neil, 2014; Barsics, 2014; Young et al., 2020; Hanley et al., 1998; Damjanovic & Hanley, 2007). In the case of new identities, in addition to being more accurate, subjects were more confident when correctly identifying a

never-encountered individual through their face than their voice. This was reflected by the fact that whereas drift rates for new faces were negative (i.e., towards a "new" response), those for new voices were positive, indicating a bias to consider voices as previously encountered individuals (i.e., a familiarity bias). Nonetheless, the drift rates for true old identities remained larger than those for new ones. Hence the subjects were attempting, despite this bias and with limited success, to correctly identify the individuals as new or old based on their voice and not just following this putative bias.

High arousal interferes with emotion-independent identity memory

Results from the Main Study point, albeit indirectly, towards a relative recognition impairment in performance, reflected in reduced accuracy and less efficiency of memory processing - as indexed by the diffusion model drift rates, argued to represent the strength of memory traces that are used to discriminate between old and new items -, of identity recognition memory for high-arousal emotions in both modalities. This possible relation was then directly tested, and confirmed, by a strong correlation between stimulus-specific recognition accuracy and the corresponding subjective arousal ratings across all emotions and modalities. We further confirmed this impairment effect in two additional experiments (Follow-up Study 1) comparing, in a within-subject design, high- and low-arousal expressions, namely Fearful vs. Sad (with Neutral recognition) and Fearful vs. Neutral (with Happy recognition). Again, results revealed a significantly reduced recognition accuracy for identities encoded with a high-arousal expression, both for faces and voices, ruling out any possible confounding effects due to potential overall performance differences among *Uni*-emotion groups from the Main Study.

Thus, the overall pattern of results across studies, measures and indices, strongly supports the conclusion of a reduced expression-independent identity recognition, when encoded in high-arousal emotions. This is consistent with the central/peripheral (or intrinsic/extrinsic) trade-off phenomenon mentioned in Introduction (for reviews, see Buchanan & Adolphs, 2002; Kensinger, 2009; Mather & Sutherland, 2011). Following this notion, we suggest that exposure to individuals expressing a high-arousal emotion enhances recognition of that specific identity-emotion combination (i.e., same stimulus), but interferes with the formation of an expression-invariant representation of the individual's identity. This could result in a worse recognition when the identity is presented in a novel expression. Notably, the arousal influence on memory was still present despite the task irrelevance of emotional expression, both during encoding (age judgment) and recognition. This finding fits well with the notion that arousal modulates selective attention towards significant or goal-relevant aspects of stimuli (Mather & Sutherland, 2011), and evidence that arousing (particularly negative) aspects of events capture attention in a relatively automatic manner (e.g., Armony, Vuilleumier, Driver, & Dolan, 2001; Dolan & Vuilleumier, 2003; Sanders et al., 2005). Moreover, these results, and their interpretation, are also consistent with studies reporting better performance in visual detection and memory tasks when in a low-arousal mood state, such as sadness (e.g., Jefferies et al., 2008; Hills et al., 2011). The proposed hypothesis would not only help explain our current findings, but also integrate them with the seemingly contradictory results in the past showing an enhanced memory for same-stimulus recognition of emotional faces or voices (e.g., Sergerie et al., 2005, Aubé et al., 2013). That is, emotional arousal would strengthen the encoding of the core features of an emotional stimulus, thus resulting in a superior recognition of the stimulus per se. However, this would be accomplished at the expense of the encoding of its "emotionally-irrelevant" features, which would then reduce the ability to identify them in a different exemplar of the same individual. This in turn, would be reflected in a difficulty to successfully generalize individuals to a novel expression, as was the case here.

Multiple exemplar memory advantage compared to repeated high-arousal expressions

Results from the *Multi* condition from the Main Study and Follow-up Study 2 collectively suggest that encoding exemplar variance help overcome the arousal-related weaker identity representation formation, discussed above, even when most (Main Study), or even all (Follow-up Study 2), of the expressions convey high arousal information. This is likely achieved through integration of different exemplars into an expression-independent representation of the individual. Indeed, as mentioned above, a significant priming effect in encoding RTs over repeated individual presentations was found in both *Uni* and *Multi* conditions, and, more importantly, the magnitude of this priming, as well as the degree of age judgment consistency, predicted recognition success during the test phase in which individuals were presented in a never-encountered-before neutral expression. It is important, however, to consider an alternative explanation, namely that this *Multi* advantage was simply due to the presence of sad stimuli in that condition in all groups. Results from the Follow-up Study 2, in which we used a reduced *Multi* condition, without this low-arousal expression, rule out this possibility, by consistently showing the advantage on recognition performance.

Results from the DDM-derived drift rates, integrating response choice and times, suggest facilitated responses towards correctly recognized *Multi* identities. A speeded response has been proposed to reflect an increased confidence in the recognition of previously encountered identities (Robinson et al., 1997; Weidemann & Kahana, 2016; Xu & Armony, 2021). Here, we confirmed this hypothesis by including an explicit measure of response confidence (*Sure/Unsure*). As expected, confident responses showed significantly shorter RTs. Furthermore, participants reported being more confident of their response when correctly recognizing *Multi* than *Uni* identities, but only for faces. This lack of significant confidence difference, compared to the DDM results, may suggest that the usage of a binary confidence rating is not precise or sensitive enough, to detect weaker or nosier difference. These findings confirm the validity of using response times as a proxy for response confidence which, in fact, may be more sensitive and efficient than implementing a binary explicit confidence rating.

Overall, our findings are consistent with previously reported memory advantage of encoding variability on identity recognition for faces (Murphy et al., 2015; Ritchie & Burton, 2017; Matthews et al., 2018) and voices (Lavan et al., 2019a; Xu & Armony, 2021). They also extend, and help generalize, those findings in several ways. First, from a methodological standpoint, they confirm that, despite some concerns on data collection and quality (reviewed by Finley & Penningroth, 2015), online memory experiments can yield meaningful results, comparable to the ones obtained from in-person studies, including those related to response times. Indeed, the effect sizes for recognition accuracy as a function of exemplar variability (*Multi* vs. *Uni*) were comparable between studies ($d = 0.17$ for voices of *Uni*-Fearful group, $d = 0.13$ for data from Xu & Armony, 2021). One of the advantages of porting an in-person study to an online platform is that it typically allows for the recruitment of larger sample sizes which, are more likely to yield more robust, statistically significant effects, as it was the case here. Nonetheless, a necessary trade-off is a reduced control over the testing environment. Additionally, our results demonstrate a strong across-modality correspondence in the effects of emotion, both in terms of facilitation and impairment, on identity memory. This goes along with the proposed similarity in voice and face identity processing (see Belin et al., 2011 for a review).

Our findings may also provide some insights into the controversial topic of eye/ear-witness reliability (Magnussen et al., 2010; Sherrin, 2016). In such cases, the suspected criminals are encountered in a few, usually only one, instances, under less-than-ideal conditions. In contrast, the recognition (line-up) test is typically conducted under tightly controlled conditions, requiring the witness to identify the target individual among a group of similar foils (with neutral expressions). Recent research suggests that encoding conditions (e.g., close/far, short/long exposure) have a substantial impact on recognition confidence and accuracy (Molinaro et al., 2021). Our results further confirm and extend this observation, showing that emotion-related variables, including expression and variability, also play a significant role in subsequent recognition accuracy and confidence.

Limitations and future directions

As is the case in many psychology studies, when conducted either online or in-person, female participants were over-represented in our sample, and we were therefore unable to explore sex differences. As some sex- and gender-based individual differences have been reported in emotion and memory studies (e.g., see Armony & Sergerie, 2007; Skuk & Schweinberger, 2013; Herlitz & Lovén, 2013; Patel et al., 2023 for memory research; see Montagne et al., 2005; Filkowski et al., 2017; Kret & de Gelder, 2012 for emotion processing), it is important in future studies to ensure enough diversity to be able to assess whether the observed effects are modulated by these factors.

Whereas we tested the encoding of in Follow-up Study 2 in three different high-arousal emotions (anger, fear and happiness/joy), we only used sadness as a low-arousal expression (as well as neutral ones in the Follow-up Study 1). This was a limitation of the datasets employed which, as most available ones, restrict the set of emotions to so-called basic ones (i.e., the ones mentioned here, plus disgust and, sometimes, surprise). Although our conclusions are supported by the direct correlation analysis between recognition accuracy and arousal scores across emotions and modalities, it would be important to further confirm our findings with other low-arousal expressions. This could be achieved by employing other emotions (e.g., contempt) and/or through morphing procedures.

We tested recognition in each modality separately, thus our results cannot directly speak to the question of emotional memory for multimodal stimuli. Nonetheless, given that we observed similar patterns for face and voice, we can speculate that these effects would also hold for audio-visual expressions, although this remains to be confirmed. Such studies would be an important contribution to the memory literature, which reports largely conflicting observations. For example, different studies have found the simultaneous presentation of face and voice stimuli help (e.g., Maguinness et al., 2021; von Kriegstein et al., 2008; Zäske et al., 2015) and hinder (e.g., Lavan et al., 2023; Cook & Wilding, 2001; Tomlin et al., 2017) voice identity learning and recognition.

Conclusion

We investigated how emotion influences face- and voice-based recognition of individuals subsequently encountered with a different expression. Specifically, our findings demonstrated that encoding identities with high-arousal expressions hindered subsequent different-expression recognition for both modalities. However, this effect could be overcome by the presentation of multiple exemplars with different expressions during encoding. In fact, these opposing effects led to a similar recognition accuracy between repeated presentation of a single low-arousal expression (e.g., sad) and multiple high-arousal ones (e.g., angry, fearful, and happy). To illustrate these findings, we can return to the movie example mentioned in the Introduction; imagine that, back in the 60's, two friends — one of whom saw *Psycho* several times while the other, in addition, watched *Touch of Evil* and *Holiday Affair* — ran into Janet Leigh when she was ordering coffee at the local diner. According to our results, the fan who saw Ms. Leigh starring in different films should have been more likely to recognize her, and therefore get her autograph.

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Declarations

Conflicts of interest The authors declare no conflicts of interest.

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